

Configuring NVMe/TCP with VMware ESXi on VSP One Block 20

v1.0

Implementation Guide

This document shows how to configure NVMe/TCP with VMware ESXi on Hitachi Virtual Storage Platform One Block 20 (VSP One B20) family storage systems. This document applies to VSP One B24, B26, and B28 storage systems.

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Preface

About this document

This guide shows how to set up and configure NVMe/TCP on a VMware ESXi system using Hitachi Virtual Storage Platform One Block 20 (VSP One B20) family storage systems. The storage system was configured using the Hitachi Command Control Interface (CCI).

The document consists of the following main sections:

- Configuring VSP One Block 20 Port Parameter settings, NVM Subsystem, and Namespace using the CCI command line interface.
- Configuring NVMe/TCP on VMware ESXi with a 100G NIC card. The steps in this section are for environments running with ESXi 8.0U3.
- Configuring the ESXi Path Selection Policy (PSP).

Intended audience

This document is intended for a technical audience who need an overview of how to configure NVMe/TCP with VMware ESXi on VSP One Block 20 family storage systems. To use this document, you must have a basic understanding of the VMware ESXi Operating System (OS), NVMe/TCP, network configuration, Hitachi Command Control Interface, and VSP One Block 20 storage systems.

Revision History

Revision	Changes	Date
v1.0	Initial Release	August 2025

Accessing product documentation

Product user documentation is available on the Hitachi Vantara Support Site: <https://docs.hitachivantara.com/>. Check this site for the most current documentation, including important updates that may have been made after the release of the product. For more information on the operating system, see the operating system documentation. For the product support matrix, see the Product Compatibility Guide: <https://compatibility.hitachivantara.com/>

For OS support, check their support site:
VMware ESXi: <https://techdocs.broadcom.com/>

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Introduction

Hitachi Virtual Storage Platform One Block 20 (VSP One B20) is a new 2U all-NVMe storage appliance with three dedicated models. All models have the same capacity (72 NVMe flash drives, the appliance, + 2x media trays) and support FC, iSCSI, and NVMe TCP connectivity. The new capabilities remove complexity such as data reduction always being on, Dynamic Drive Protection (DDP) removing complicated RAID setups, and Dynamic Carbon Reduction delivering real-world reductions in power consumption.

- VSP One Block 24 – 256GB Cache + SW Advanced Data Reduction (ADR) + 24 cores
- VSP One Block 26 – 768GB Cache + 2x Compression Accelerator Module (CAM) + 24 cores
- VSP One Block 28 – 1TB Cache + 4x CAM + 64 cores

This document describes the configuration of NVMe/TCP with VMware ESXi OS using VSP One Block 20 with VMware High Performance Plugin (HPP) with the help of the Hitachi Command Control Interface (CCI).

Purpose

This document describes the procedure for configuring NVMe/TCP on VMware ESXi. It focuses on the configuration process for VMware ESXi 8.0U3 with VSP One Block 20 using the NVMe/TCP protocol.

The purpose of this guide is to provide an overview of the implementation that includes VSP One Block 20 with NVMe/TCP in an ESXi Host environment. In addition, this document includes NVMe/TCP port configuration, Subsystem ID creation, NVMe namespace provisioning, and host-side configuration for storage namespace detection in a VMware setup.

NVMe/TCP on VMware

NVMe/TCP, also known as NVMe over TCP, is a storage networking protocol that extends the NVMe protocol across TCP/IP networks. In the context of VMware, NVMe/TCP enables the use of NVMe storage devices over the standard Ethernet networks without requiring specialized hardware such as Remote Direct Memory Access (RDMA).

In VMware ESXi environments, support for NVMe over TCP allows virtual machines to access NVMe storage devices over the network with improved performance and reduced latency.

Before you begin, see the following for more information:

- For NVMe/TCP support specifics, see the OS release notes, related installation documents, and storage configuration guides.
- For additional details about NVMe/TCP support, see the NIC user guides.

NVMe/TCP Connectivity with High Performance Plugin

To connect NVMe/TCP, install a supported dual-port 100G NIC on a server running VMware ESXi 8.0U3 OS. The NIC ports connect to VSP One Block 20 ports through two 100G Network Switches.

The High Performance Plugin (HPP) in VMware ESXi optimizes the handling of NVMe devices, ensuring that they deliver the best possible performance. With HPP installed, a server communicates with a storage LUN through multiple physical paths between the Host NIC in the server and the device storage controllers. This document describes how to configure the NVMe/TCP on VMware ESXi 8.0U3 with the HPP.

Connectivity Diagram

The following diagram shows Port 3A and Port 4A of a VSP One Block 26 storage system connected to an ESXi host through two 100G network switches:

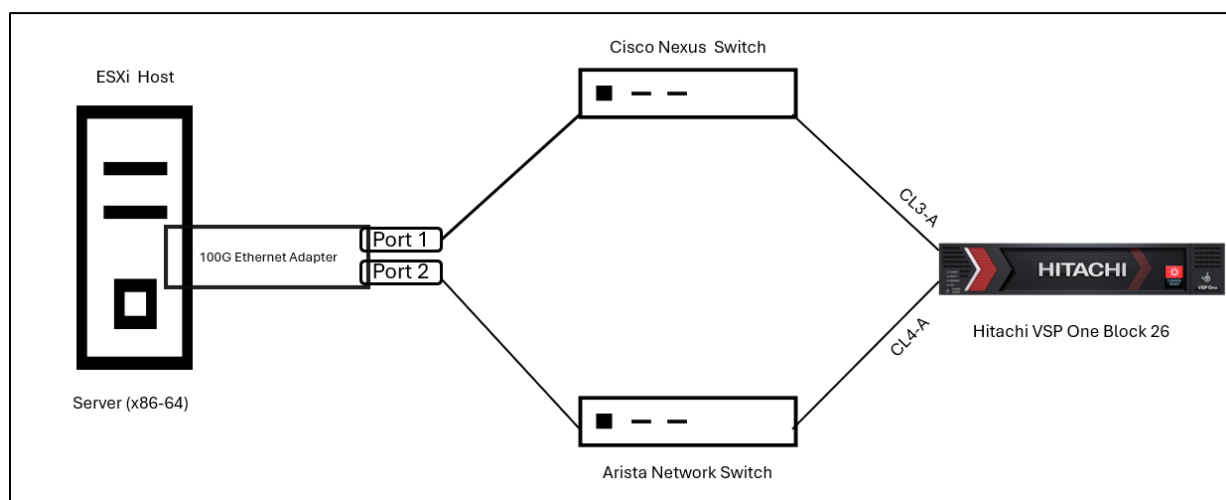


Figure 1: Port 3A and Port 4A of a VSP One Block 26 storage system connected to ESXi host through two 100G network switches

The following diagram shows the workflow of NVMe/TCP where the host NIC communicates with the NVM Subsystem to access the namespaces created from VSP One Block 20.

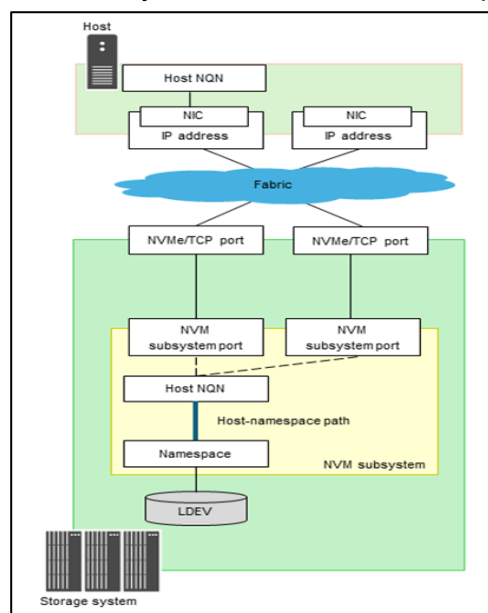


Figure 2: Workflow of NVMe/TCP

Hardware Requirements

The following lists the hardware used in this setup:

- NVMe/TCP host system: Hitachi Advanced Server HA820 G3
- Host NIC: Mellanox MCX623106AN-CDAT Dual Port 100Gb/s QSFP56
- CCI server: HPE ProLiant DL380 Gen 10
- Switches: Cisco 93180YC-FX3, Arista 7050SX3-48YC12-F
- Storage system: Hitachi VSP One Block 26 (SVOS 10.2.1)

Software Requirements

The following lists the software used in this setup:

- OS install media: ESXi 8.0U3
- ESXi OS build number: 24859861 (selected to enhance performance related to SQ Flow Control settings)
- vCenter version: 8.0.0.10000
- CCI software: Command Control Interface version 01-76-02/02

Getting Ready

1. Complete the connectivity by installing an NVMe-supported dual port NIC into a compatible server. The NIC ports are connected to VSP One Block 20 ports through two 100G network switches.

2. Install ESXi 8.0U3 locally.

3. Install the supported ethernet adapter on the ESXi host.

4. To generate an NVMe Qualified Name (NQN) for the adapter, run the following command:

```
esxcfg-module -s 'vmknvme_hostnqn_format=0' vmknvme
```

5. For changes done in step 4 to take effect, reboot the ESXi host.

6. Note the Host NQN information by running the following command. The output will be similar to the following output:

```
esxcli nvme info get  
Host NQN: nqn.2014-08.org.nvmexpress:uuid:662612c9-f672-7621-6074-b83fd22a0b10
```


Note: The Host adapter NQN does not change with host restarts or driver updates; however, it changes with a new host OS installation.

Best Practices

In ESXi 8.0U3f and earlier builds, certain operations on the ESXi host may experience slower performance, such as:

- Delays during virtual machine creation
- Delays during datastore creation
- Slower disk allocation within VMs

These delays may be associated with the default configuration of SQ Flow Control, which is enabled by default on ESXi hosts. To optimize performance, Broadcom recommends upgrading to ESXi 8.0U3g (build 24859861) and disabling SQ Flow Control by completing the following steps:

1. To disable SQ Flow Control, run the following command:

```
# esxcli system module parameters set -m vmknvme -p
"vmknvme_disable_sq_flow_control=0"
```

If any other parameter is already configured and there is a need to set `vmknvme_disable_sq_flow_control` while preserving the existing settings, ensure that all parameters are specified in the command. Use the following format:

```
esxcli system module parameters set -m <module name> -p "<p1>=<v1>
<p2>=<v2> <p3>=<v3>... <pn>=<vn>"
```

The following command is applied to retain the disabled state for both `'vmknvme_hostnqn_format'` and `'vmknvme_disable_sq_flow_control'`.

```
# esxcli system module parameters set -m vmknvme -p
"vmknvme_hostnqn_format=0 vmknvme_sq_flow_control_enabled=0"
```

2. To confirm that SQ Flow Control is successfully disabled, run the following command:

```
# esxcli system module parameters list -m vmknvme
```

```
[root@localhost:~] esxcli system module parameters list -m vmknvme
Name      Type  Value  Description
-----
vmknvme_adapter_num_cmpl_queues  uint  Number of FSA completion queues for NVMe-oF adapter, min: 1, max: 16, default: 4
vmknvme_bind_intr               uint  If enabled, the interrupt cookies are binded to completion worlds. This parameter is only applied when using driver completion worlds.
vmknvme_cmpl_world_type         uint  completion world type, FSA: 0, VMKNVME: 1
vmknvme_ctrl_recover_initial_attempts  uint  Number of initial controller recover attempts, MIN: 2, MAX: 30
vmknvme_ctrl_recover_method     uint  controller recover method after initial recover attempts, RETRY: 0, DELETE: 1
vmknvme_cw_rate                 uint  Number of completion worlds per IO queue (NVMe/PCIe only). Number is a power of 2. Applies when number of queues less than 4.
vmknvme_enable_noioab          uint  If enabled, driver will split the commands based on NOIOS.
vmknvme_hostnqn_format          uint  0  HostNQN format, UUID: 0, HostName: 1
vmknvme_io_queue_num           uint  vmknvme IO queue number for NVMe/PCIe adapter: pow of 2 in [1, 16]
vmknvme_io_queue_size          uint  IO queue size: [8, 1024]
vmknvme_iosplit_workaround     uint  If enabled, qdepth in FSA layer is half size of vmknvme settings.
vmknvme_log_level              uint  log level: [0, 20]
vmknvme_max_rrp_entries_num    uint  User defined maximum number of RRP entries per controller; default value is 0
vmknvme_sq_flow_control_enabled  uint  0  Enable/disable SQ flow control. Enabled: 1, disabled: 0, default: 1
vmknvme_stats                  uint  NVme statistics per controller (NVMe/PCIe only now). Logical OR of flags for collecting. 0x0 for disable, 0x1 for basic data (IO pattern)
, 0x2 for histogram without IO block size, 0x4 for histogram with IO block size. Default 0x2.
vmknvme_total_io_queue_size     uint  Aggregated IO queue size of a controller, MIN: 64, MAX: 4096
vmknvme_use_default_domain_name  uint  If set to 1, the default domain name "com.vmware", not the system domain name will always be used to generate host NQN. Not used: 0, used
: 1, default: 0
[root@localhost:~]
```

3. Reboot the system.

Configuring the VSP One Block 20 NVM Subsystem

This section describes how to configure the VSP One Block 20 NVM Subsystem with CCI.

Before you begin:

Confirm the storage system ESM IP address and configure the 'horcm' file on a CCI management server to start the 'horcm' instance. For detailed information on CCI installation, configuration, and commands, see the Command Control Interface manuals. CCI server must be running version 01-75-03/03 or later.

The following example shows a CCI configuration file used on VMware ESXi with Command Control Interface software:

```
# cat horcm6824.conf
HORCM_MON
#ip address      service      poll(10ms)  timeout(10ms)
127.0.0.1        horcm6824    1000        3000

HORCM_CMD
\\.\IPCMD-172.23.68.25-31001

# cat /etc/services |grep horcm6824
horcm6824        485236/udp      #horcm6824

# horcmstart.sh 6824
HORCM inst 6824 has already been running.
```

Storage Port Settings

This section describes how to configure the 100G TCP storage port.

Verify NVMe mode settings for the storage ports (3A and 4A) by completing the following steps:

1. Verify that the port mode is 'NVME_TCP'.
2. Verify the port mode using the following CCI raidcom command:

```
# raidcom get port -key detail -I6824
```

PORT	TYPE	ATTR	SPD	LPID	FAB	CONN	SSW	SL	Serial#	WWN	PHY_PORT	PORT_MODE	RSID	SECURITY
CL3-A	NVME_TCP	TAR	100G	00	N	UNKN	N	0	8xxxx	-	-	NVME_TCP	0	-
CL4-A	NVME_TCP	TAR	100G	00	N	UNKN	N	0	8xxxx	-	-	NVME_TCP	0	-

By default, the port mode of 100G CHB is NVME_TCP because the default support protocol is NVMe/TCP.

Configuring Topology Settings

Configure the 100G TCP storage ports by completing the following steps:

1. Assign network IP addresses to the 100G TCP storage ports in different subnet ranges, as shown in the following example.

In the following example, network IP addresses are assigned against CL3-A and CL4-A 100G storage ports.

Assign the network IP addresses to the 100G TCP storage ports and host NIC ports in the same range of IP addresses in two different subnets.

```
# raidcom modify port -port cl3-A -ipv4_address 192.168.10.10 -ipv4_subnetmask 255.255.255.0 -I6824
```

```
# raidcom modify port -port cl3-A -mtu 9000 -window_size 2048k -I6824
```

```
# raidcom modify port -port cl4-A -ipv4_address 192.168.11.10 -ipv4_subnetmask 255.255.255.0 -I6824
```

```
# raidcom modify port -port cl4-A -mtu 9000 -window_size 2048k -I6824
```

2. Verify the IP address configuration by running the following command:

```
# raidcom get port -port cl4-A -key opt -I6824
PORT : CL4-A
TCP_OPT : IPV6_D : SACK_E : DACK_E : INS_D : VTAG_D
TCP_MTU : 9000
WSZ : 2048KB
KA_TIMER : 60
TCP_PORT_IO_CONTROLLER : 4420
TCP_PORT_DISCOVERY_CONTROLLER : 8009
IPV4_ADDR : 192.168.11.10
IPV4_SMSK : 255.255.255.0
IPV4_GWAD : 0.0.0.0
IPV6_ADDR_INF : INV : AM : fe80::
IPV6_GADR_INF : INV : AM : ::
IPV6_GWAD_INF : INV : :: : ::
VLAN_ID : -
VP_MODE : D
IPV6_GADR2_INF : INV : AM : ::
MAC_ADDR : f8:48:97:47:e3:95
```

```
# raidcom get port -port cl3-A -key opt -I6824
PORT : CL3-A
TCP_OPT : IPV6_D : SACK_E : DACK_E : INS_D : VTAG_D
TCP_MTU : 9000
WSZ : 2048KB
KA_TIMER : 60
TCP_PORT_IO_CONTROLLER : 4420
TCP_PORT_DISCOVERY_CONTROLLER : 8009
IPV4_ADDR : 192.168.10.10
IPV4_SMSK : 255.255.255.0
IPV4_GWAD : 0.0.0.0
IPV6_ADDR_INF : INV : AM : fe80::
IPV6_GADR_INF : INV : AM : ::
IPV6_GWAD_INF : INV : :: : ::
```

```
VLAN_ID : -
VP_MODE : D
IPV6_GADDR2_INF : INV : AM : ::
MAC_ADDR : f8:48:97:47:dc:0b
```

Configuring the NVM Subsystem

Create an NVM Subsystem and register the 100G TCP storage ports to the NVM Subsystem by completing the following steps:

1. Create an NVM Subsystem on the storage system with the CCI raidcom command.
2. Verify whether there is an existing NVM Subsystem and then create the NVM Subsystem with an available ID.

In the following example, the NVM Subsystem is created with ID 5:

```
# raidcom add nvm_subsystem -nvm_subsystem_id 5 -nvm_subsystem_name
Mellanox_100GTCNVMe -host_mode VMWARE_EX -request_id auto -I6824
```

```
# raidcom get nvm_subsystem -nvm_subsystem_id 5 -I6824
```

NVMSS_ID	RGID	NVMSS_NAME	SECURITY	T10PI	HMD	HMO_BITS
5	0	Mellanox_100GTCNVMe	ENABLE	DISABLE	VMWARE_EX	-

```
# raidcom get nvm_subsystem -key opt -IH6824
```

NVMSS_ID	NVMSS_NAME	NVMSS_NQN
5	Mellanox_100GTCNVMe	nqn.1994-04.jp.co.hitachi:nvme:storage-subsystem-sn.8-10004-nvmssid.00005

3. Register the 100G TCP storage ports with the NVM Subsystem using the CCI raidcom command.

In the following example, storage ports CL3-A and CL4-A are registered with NVM Subsystem ID 5:

```
# raidcom add nvm_subsystem_port -nvm_subsystem_id 5 -port CL3-A -request_id auto
-I6824
```

```
# raidcom add nvm_subsystem_port -nvm_subsystem_id 5 -port CL4-A -request_id auto
-I6824
```

```
# raidcom get nvm_subsystem_port -nvm_subsystem_id 5 -I6824
```

PORT	NVMSS_ID	NVMSS_NAME
CL3-A	5	Mellanox_100GTCNVMe
CL4-A	5	Mellanox_100GTCNVMe

Note:

Hitachi's NVMe/TCP storage system supports exporting multiple namespaces to an ESXi host through a single subsystem. To optimize stability and resource usage, VMware recommends limiting the ESXi host connections to a maximum of eight subsystems, regardless of the number of required namespaces. This prevents resource exhaustion and prevents controllers going offline if a target port becomes unreachable or offline. For more information, see:

<https://knowledge.broadcom.com/external/article/378512/recommendation-for-esxi-nvmetcp-configur.html>

Updating Namespace Security Settings

Update the Namespace security settings by completing the following steps:

1. Make a note of the Host NQN to use when configuring the NVM Subsystem. To check the Host NQN on a VMware ESXi system, run the following command:

```
# esxcli nvme info get
Host NQN: nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
```

2. Set the Host NQN to allow access to the NVM Subsystem with the CCI raidcom command, as shown in the following example:

```
# raidcom add host_nqn -nvm_subsystem_id 5 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824
```

```
# raidcom get host_nqn -nvm_subsystem_id 5 -I6824
```

NVMSS_ID	NVMSS_NAME	HOST_NQN
5	Mellanox_100GTCPNVMe	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10

Creating a Namespace

Create a Namespace by registering the LDEV with the NVM Subsystem. A Namespace is then created for that LDEV with a unique Namespace ID.

1. Create new LDEVs with the CCI raidcom command, as shown in the following example:

```
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:00 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:01 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:02 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:03 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:04 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:05 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:06 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:07 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:08 -capacity 10G -I6824
raidcom add ldev -parity_grp_id 1-1 -ldev_id 00:09 -capacity 10G -I6824
```

2. Format the newly created LDEVs using the raidcom command, as shown in the following example:

```
raidcom initialize ldev -operation fmt -ldev_id 00:00 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:01 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:02 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:03 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:04 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:05 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:06 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:07 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:08 -I6824
raidcom initialize ldev -operation fmt -ldev_id 00:09 -I6824
```

3. Create the Namespace for the LDEVs by registering them with the Storage Subsystem ID, as shown in the following example:

```
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:00 -request_id auto -I6824
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:01 -request_id auto -I6824
```

```

raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:02 -request_id auto -I6824
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:03 -request_id auto -I6824
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:04 -request_id auto -I6824
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:05 -request_id auto -I6824
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:06 -request_id auto -I6824
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:07 -request_id auto -I6824
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:08 -request_id auto -I6824
raidcom add namespace -nvm_subsystem_id 5 -ns_id auto -ldev_id 00:09 -request_id auto -I6824

```

4. Verify that the Namespaces are created for the LDEVs in the storage subsystem:

```
# raidcom get namespace -nvm_subsystem_id 5 -I6824
```

NVMSS_ID	NVMSS_NAME	NSID	LDEVID	CAPACITY (BLK)
5	Mellanox_100GTCNVMe	1	0	20971520
5	Mellanox_100GTCNVMe	2	1	20971520
5	Mellanox_100GTCNVMe	3	2	20971520
5	Mellanox_100GTCNVMe	4	3	20971520
5	Mellanox_100GTCNVMe	5	4	20971520
5	Mellanox_100GTCNVMe	6	5	20971520
5	Mellanox_100GTCNVMe	7	6	20971520
5	Mellanox_100GTCNVMe	8	7	20971520
5	Mellanox_100GTCNVMe	9	8	20971520
5	Mellanox_100GTCNVMe	10	9	20971520

Configuring Namespace Path Settings

To configure the Namespace path settings, complete the following steps:

1. Register the Host NQN to the Namespace ID to allow access for the Host.
2. Set the Host NQN-Namespace Path with the CCI Namespace Path configuration command, as shown in the following example:

```

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 1 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 2 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 3 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 4 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 5 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 6 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 7 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 8 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 9 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

raidcom add namespace_path -nvm_subsystem_id 5 -ns_id 10 -host_nqn nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10 -request_id auto -I6824

```

3. Verify the Namespace path configuration by running the raidcom command, as shown in the following example:

```
# raidcom get namespace_path -nvm_subsystem_id 5 -I6824
```

NVMSS_ID	NVMSS_NAME	NSID	LDEV#	HOST_NQN
5	Mellanox_100GTCNVMe	1	0	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	2	1	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	3	2	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	4	3	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	5	4	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	6	5	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	7	6	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	8	7	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	9	8	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10
5	Mellanox_100GTCNVMe	10	9	nqn.2014-08.org.nvmexpress:uuid:664b5717-47ff-5601-3aee-b83fd22a0b10

Configuring NVMe/TCP on a VMware ESXi Host for VSP One Block 20 Components

This section describes how to configure the ethernet adapter on the ESXi host, which involves setting up VMkernel binding for a TCP network adapter, adding a software adapter for NVMe over TCP, and then adding the NVMe controller.

Before you begin

- Install the OS by following the instructions in the vendor installation documents.
- Verify that the NVMe/TCP NICs are detected by the OS inbox driver.

If required, install the latest NVMe/TCP supported driver provided by the NIC product vendor.

Configuring NVMe/TCP on the ESXi Host

Configuring NVMe/TCP on the ESXi Host consists of the following high-level steps. Ensure that an adapter that supports NVMe over TCP technology is installed on the ESXi host.

- Configuring VMkernel Binding for the NVMe over TCP Adapter
- Adding the software NVMe over TCP Adapter
- Adding target controllers for NVMe over TCP

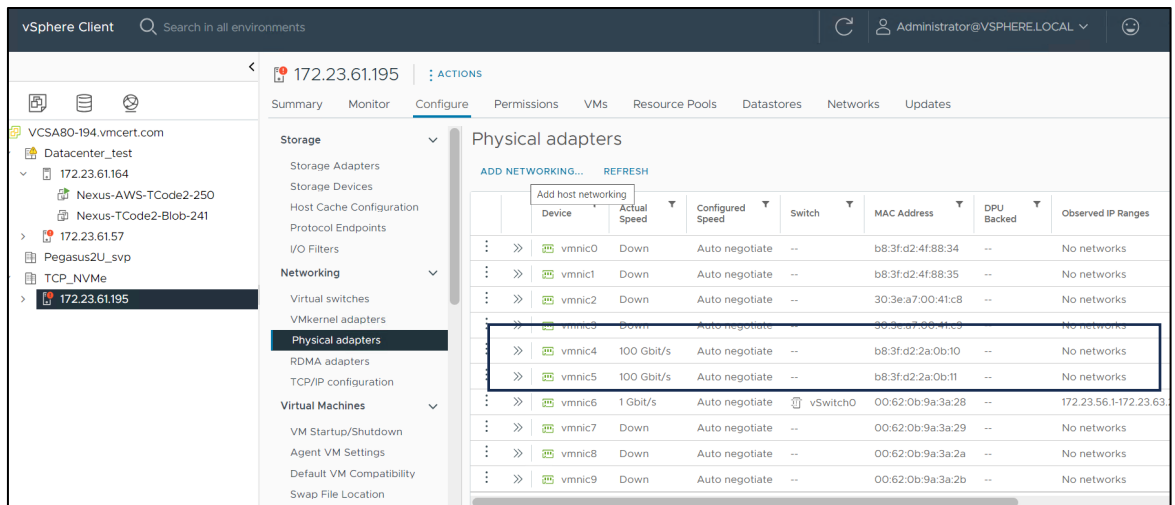
Configuring VMkernel Binding for the NVMe over TCP Adapter

Port binding for NVMe over TCP involves creating a virtual switch and connecting the physical network adapter and the VMkernel adapter to the virtual switch.

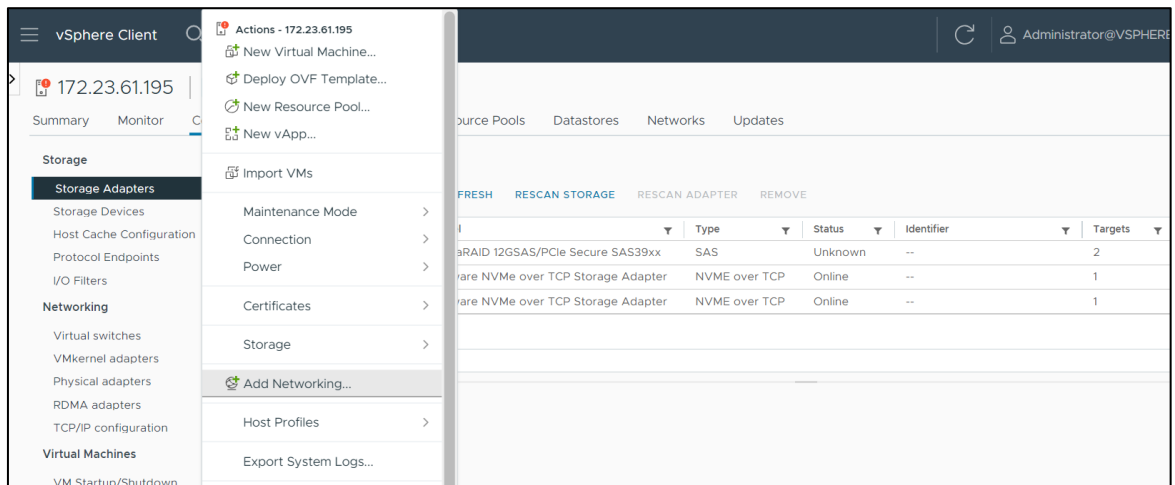
You can configure VMkernel binding for the TCP adapter using a vSphere standard switch with one uplink per switch. Configuring the network connection involves creating a virtual VMkernel adapter for each physical network adapter.

To configure VMkernel Binding for the TCP Adapter with a vSphere Standard Switch, perform the following steps:

1. From vSphere Client, select **Configure > Networking > Physical adapters** and identify the 100G vmnic devices.

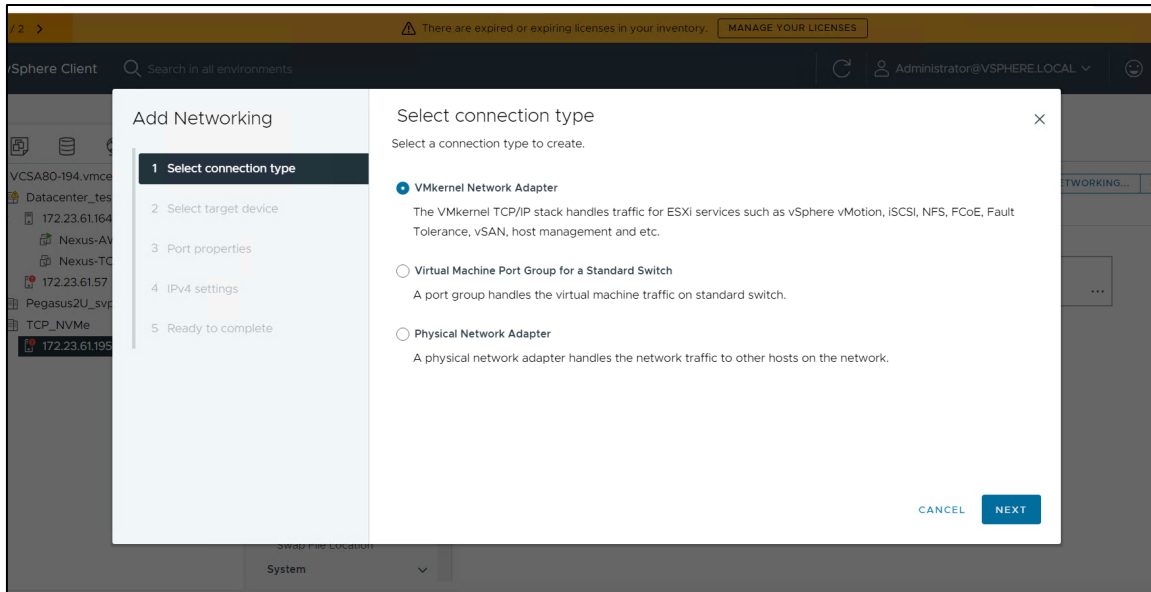


2. Select the host and then select **Actions > Add Networking**.

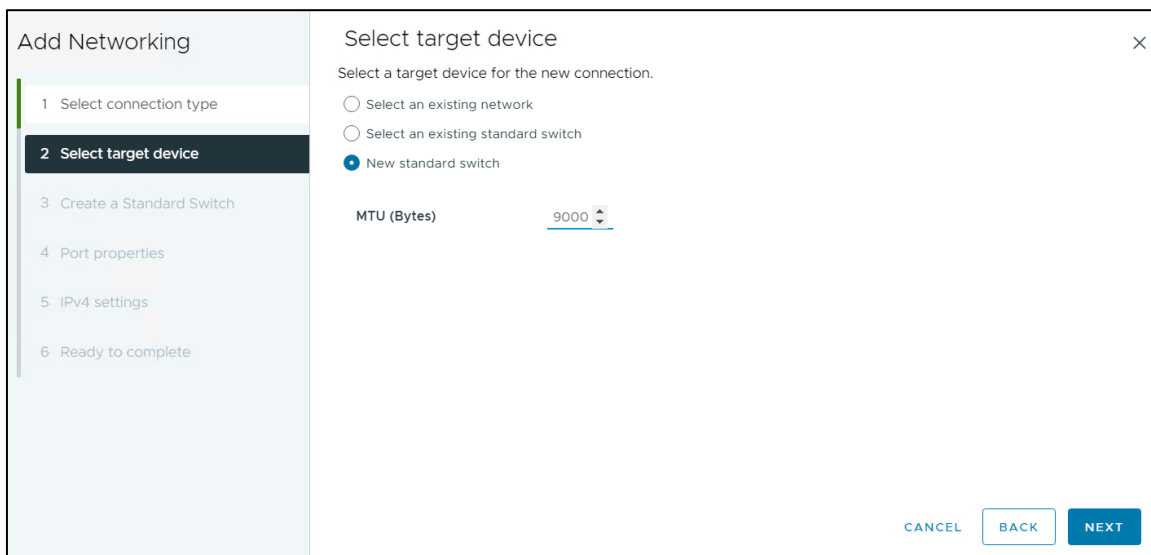


3. Select **VMkernel Network Adapter** and click **NEXT**.

In this scenario, we configured VMkernel Network Adapter 1 with a vSphere Standard Switch. You must configure VMkernel Network Adapter 2 and another vSphere Standard Switch the same way.



4. Select **New standard switch**, set the MTU value to 9000, and then click **NEXT**.



5. Navigate to the Create a Standard Switch tab and create a standard switch by shifting the vmnic from the Unclaimed adapters list to the Active adapters section.

Add Networking

1 Select connection type
2 Select target device
3 Create a Standard Switch
4 Port properties
5 IPv4 settings
6 Ready to complete

Create a Standard Switch

Assign free physical network adapters to the new switch.
Select unclaimed (free) adapters on the host and move them to the switch.

MOVE UP MOVE DOWN

Unclaimed adapters

vmnic1

vmnic2

vmnic5

vmnic6

vmnic7

vmnic8

vmnic9

Active adapters

(New) vmnic4

Standby adapters

Unused adapters

All Properties CDP

Properties

Adapter

Name

Location

Driver

Status

Status

Actual speed, Duplex

Configured speed, Duplex

Networks

Network I/O Control

Status

- In the Port properties tab, select the IP settings as **IPv4**, and enable the NVMe over TCP service from the Available services list.

Add Networking

1 Select connection type
2 Select target device
3 Create a Standard Switch
4 Port properties
5 IPv4 settings
6 Ready to complete

Port properties

Specify VMkernel port settings.

Network label

vSwitch1

VLAN ID

None (0)

IP settings

IPv4

MTU

Get MTU from switch

9000

TCP/IP stack

Default

Available services

Enabled services

☐ vMotion

☐ Provisioning

☐ Fault Tolerance logging

☐ Management

☐ vSphere Replication

☐ vSphere Replication NFC

☐ vSAN

☐ vSphere Backup NFC

☒ NVMe over TCP

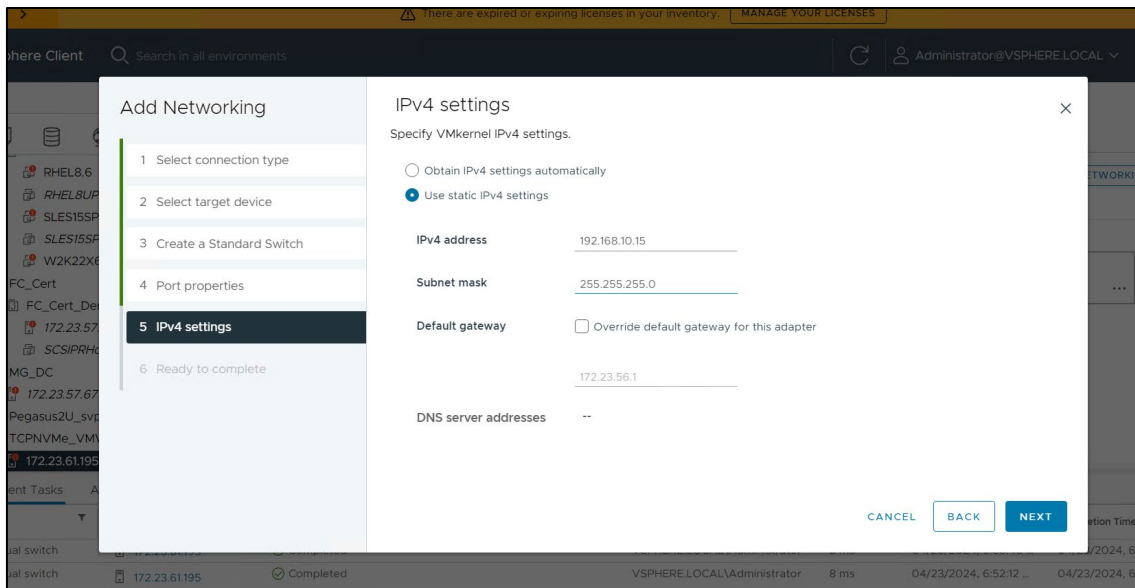
☐ NVMe over RDMA

CANCEL

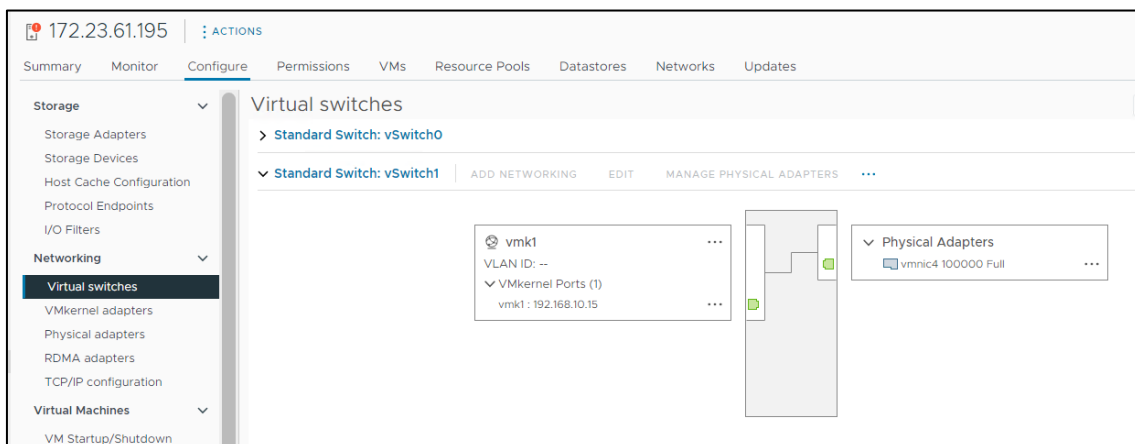
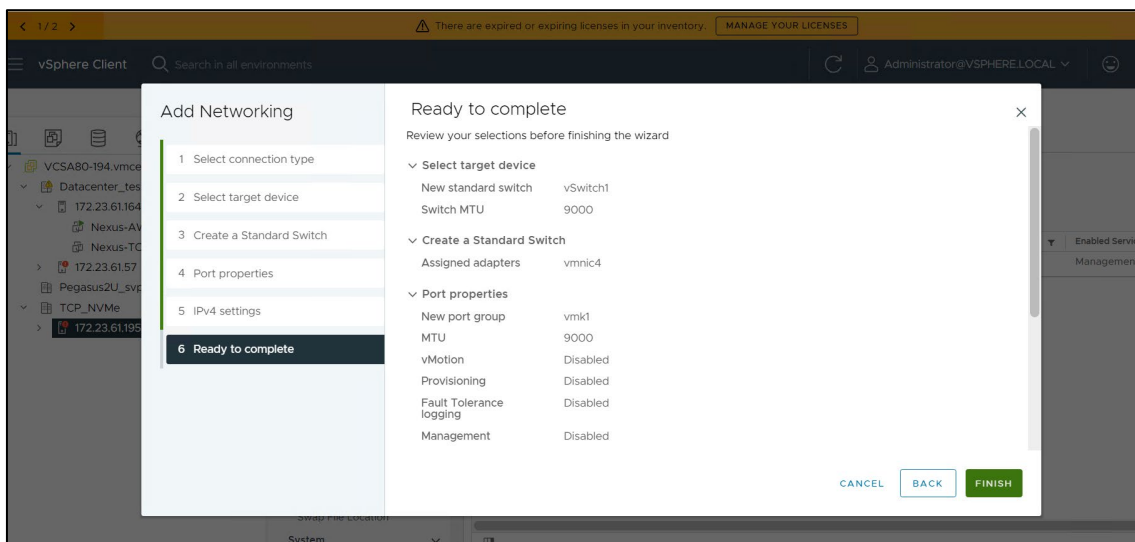
BACK

NEXT

- In the IPv4 settings tab, select **Use static IPv4 settings** and add a private IP for VMkernel adapter 1 according to the storage subnet range.

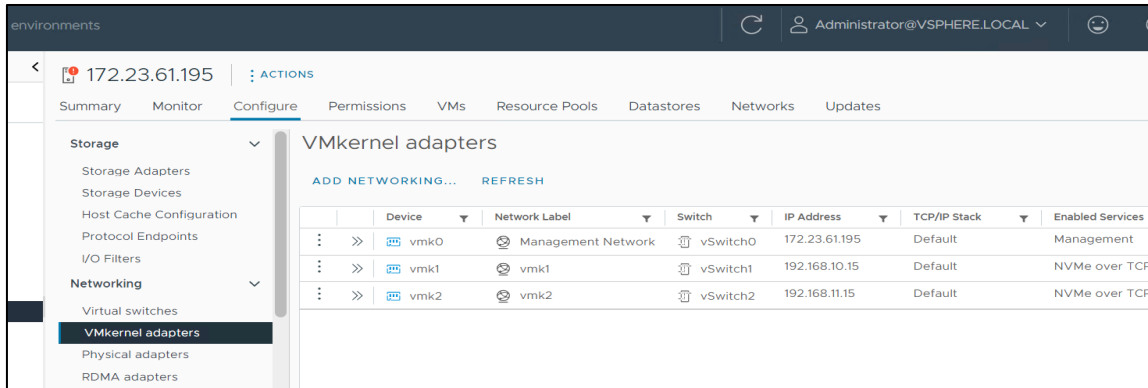


8. Verify that your switch is correctly configured.



- Configure both the VMkernels and switches, and verify that the details of both the VMkernel adapters are visible.

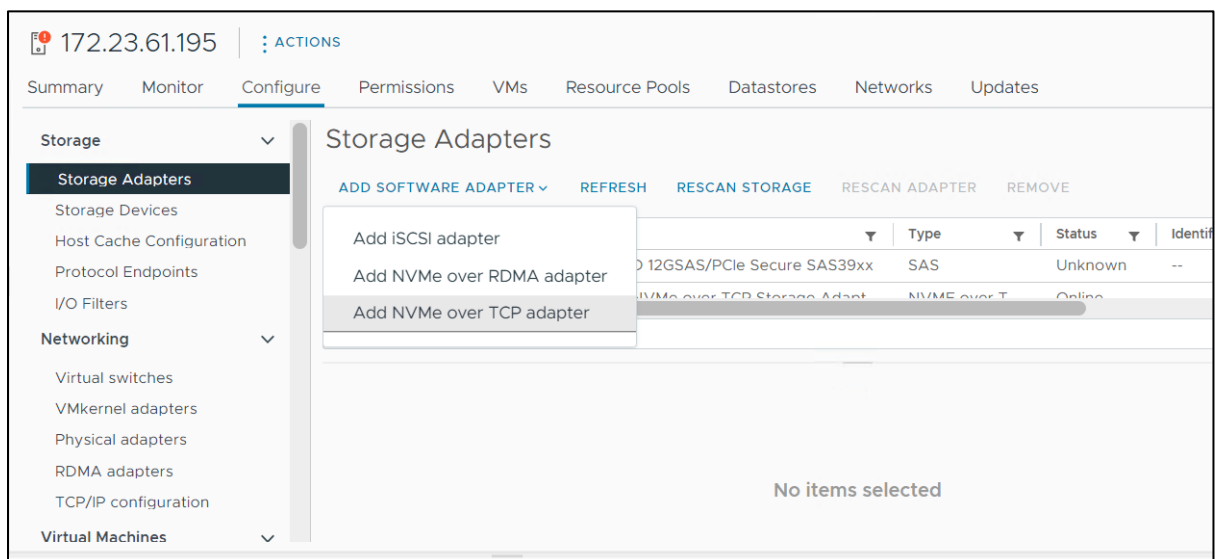
In the following screenshot, two private IPs, 192.168.10.15 and 192.168.11.15, according to their storage subnet ranges, are assigned to the VMkernel adapters vmk1 and vmk2, and vSwitches vSwitch1 and vSwitch2.



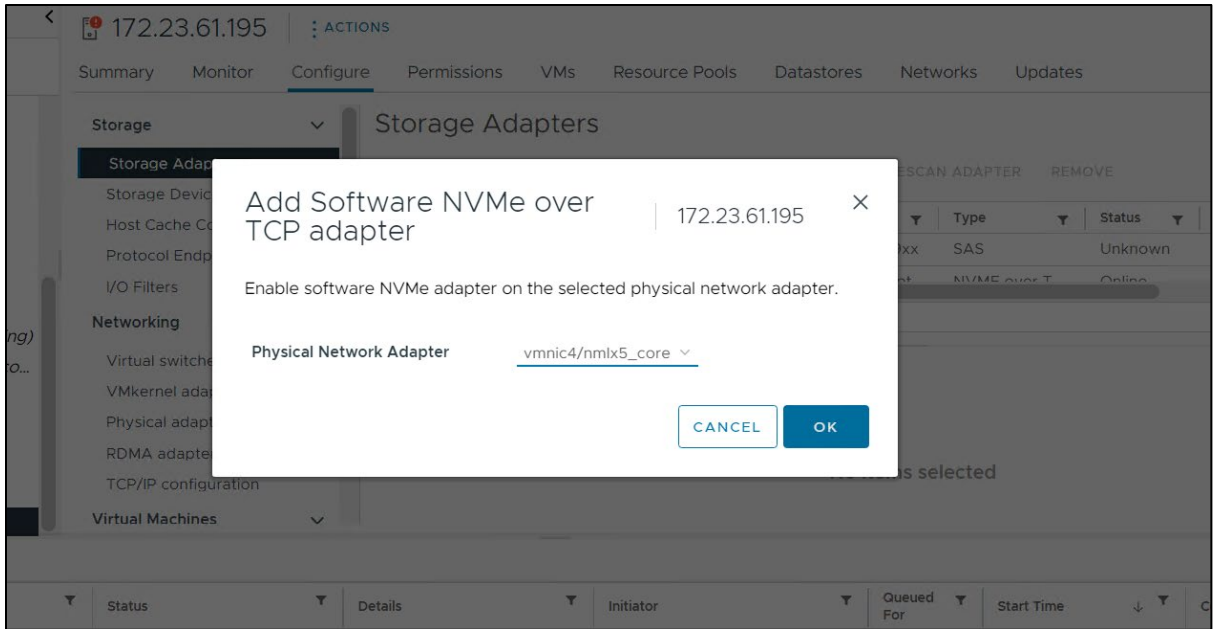
Adding the software NVMe over TCP Adapter

ESXi supports NVMe over TCP software adapters. To add the software NVMe over TCP adapter, perform the following steps:

- From the vSphere Client, navigate to the ESXi host and select the **Configure** tab.
- Under Storage, select **Storage Adapters**, select the **Add Software Adapter** drop-down menu, and then select **Add NVMe over TCP adapter**.



- Select a TCP network adapter (vmnic) from the drop-down menu while adding software NVMe and click **OK**.

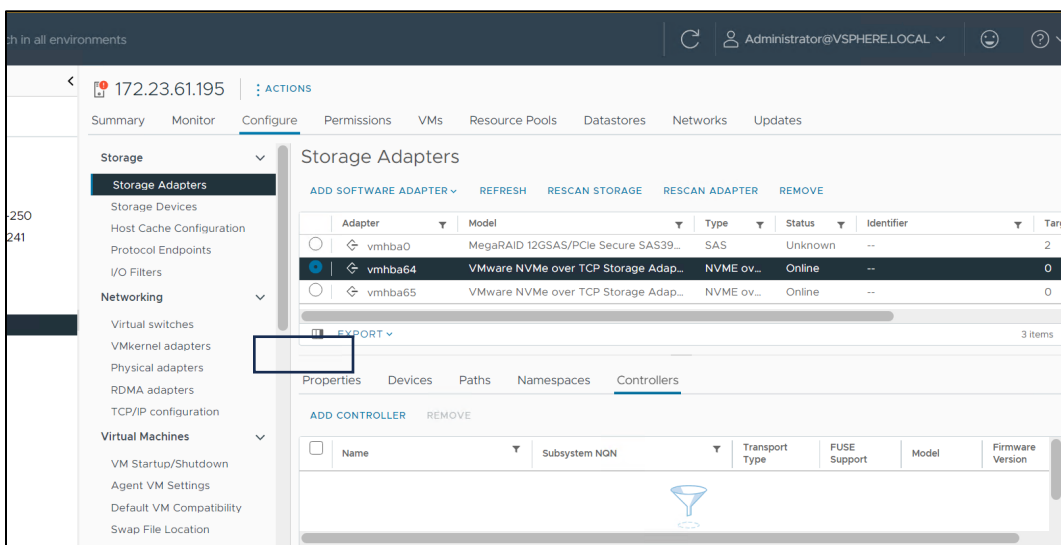


Adding target controllers for NVMe over TCP

Use the vSphere Client to add an NVMe controller. After adding the controller, the NVMe namespaces associated with the controller become available to the ESXi host. The NVMe storage devices that represent the namespaces in the ESXi environment appear on the storage devices list.

To add the controllers for NVMe over TCP adapters, complete the following steps:

1. From the vSphere Client, navigate to the ESXi host and select the **Configure** tab.
2. Under Storage, select **Storage Adapters**, and select the adapter (vmhba64) to configure.
3. Select the **Controllers** tab and click **Add Controller**.



- In the Add controller dialog box, select **Manually**, and enter the Subsystem NQN, Storage port IP, and Port Number, and then click **OK**.

The controller must be added as follows:

Adapter	Model	Type	Status	Identifier	Target
vmhba0	MegaRAID 12GSAS/PCIe Secure SAS39...	SAS	Unknown	--	2
vmhba64	VMware NVMe over TCP Storage Adap...	NVME ov...	Online	--	1
vmhba65	VMware NVMe over TCP Storage Adap...	NVME ov...	Online	--	0

Name	Subsystem NQN	Transport Type	FUSE Support	Model
nqn.1994-04.jp.co.hitachi:nvme:sto...	nqn.1994-04.jp.co.hitachi:nvme:sto...	tcp	true	HITACHI SVOS-RF-Syste...

The status of the paths is shown as Active and Active (I/O):

Runtime Name	Target	LUN	Status
vmhba64:CO:TO:L1		1	Active
vmhba64:CO:TO:L2		2	Active
vmhba64:CO:TO:L3		3	Active
vmhba64:CO:TO:L4		4	Active

Storage Adapters

Adapter	Model	Type	Status	Identifier
vmhba0	MegaRAID 12GSAS/PCIe Secure SAS39xx	SAS	Unknown	--
vmhba64	VMware NVMe over TCP Storage Adapt...	NVME over T...	Online	--
vmhba65	VMware NVMe over TCP Storage Adapt...	NVME over T...	Online	--

EXPORT

Properties Devices Paths Namespaces Controllers

ENABLE DISABLE

Runtime Name	Target	LUN	Status
vmhba64:C0:T0:L0		0	Active (I/O)
vmhba64:C0:T0:L1		1	Active (I/O)
vmhba64:C0:T0:L2		2	Active (I/O)
vmhba64:C0:T0:L3		3	Active (I/O)

When the controller is added on the ESXi host, namespaces associated with the controller must appear in the ESXi host as follows:

Storage Adapters

ADD SOFTWARE ADAPTER REFRESH RESCAN STORAGE RESCAN ADAPTER REMOVE

Adapter	Model	Type	Status	Identifier
vmhba0	MegaRAID 12GSAS/PCIe Secure SAS39xx	SAS	Unknown	--
vmhba64	VMware NVMe over TCP Storage Adapt...	NVME over T...	Online	--
vmhba65	VMware NVMe over TCP Storage Adapt...	NVME over T...	Online	--

EXPORT

Properties Devices Paths Namespaces Controllers

Name	Capacity
eui.50802714000000000060e82827140009	10 GB
eui.50802714000000000060e82827140005	10 GB
eui.50802714000000000060e82827140006	10 GB
eui.50802714000000000060e82827140007	10 GB
eui.50802714000000000060e82827140008	10 GB

Additionally, you can verify the list of Namespaces on the ESXi host using the following command:

```
# esxcfg-mpath -b
```

```
eui.50802714000000000060e82827140000 : NVMe TCP Disk (eui.50802714000000000060e82827140000)
  vmhba64:C0:T0:L0 LUN:0 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L0 LUN:0 state:active Local HBA vmhba65 channel 0 target 0

eui.50802714000000000060e82827140001 : NVMe TCP Disk (eui.50802714000000000060e82827140001)
  vmhba64:C0:T0:L1 LUN:1 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L1 LUN:1 state:active Local HBA vmhba65 channel 0 target 0

eui.50802714000000000060e82827140002 : NVMe TCP Disk (eui.50802714000000000060e82827140002)
  vmhba64:C0:T0:L2 LUN:2 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L2 LUN:2 state:active Local HBA vmhba65 channel 0 target 0

eui.50802714000000000060e82827140003 : NVMe TCP Disk (eui.50802714000000000060e82827140003)
  vmhba64:C0:T0:L3 LUN:3 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L3 LUN:3 state:active Local HBA vmhba65 channel 0 target 0

eui.50802714000000000060e82827140004 : NVMe TCP Disk (eui.50802714000000000060e82827140004)
  vmhba64:C0:T0:L4 LUN:4 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L4 LUN:4 state:active Local HBA vmhba65 channel 0 target 0
```



```
eui.50802714000000000060e82827140005 : NVMe TCP Disk (eui.50802714000000000060e82827140005)
  vmhba64:C0:T0:L5 LUN:5 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L5 LUN:5 state:active Local HBA vmhba65 channel 0 target 0

eui.50802714000000000060e82827140006 : NVMe TCP Disk (eui.50802714000000000060e82827140006)
  vmhba64:C0:T0:L6 LUN:6 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L6 LUN:6 state:active Local HBA vmhba65 channel 0 target 0

eui.50802714000000000060e82827140007 : NVMe TCP Disk (eui.50802714000000000060e82827140007)
  vmhba64:C0:T0:L7 LUN:7 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L7 LUN:7 state:active Local HBA vmhba65 channel 0 target 0

eui.50802714000000000060e82827140008 : NVMe TCP Disk (eui.50802714000000000060e82827140008)
  vmhba64:C0:T0:L8 LUN:8 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L8 LUN:8 state:active Local HBA vmhba65 channel 0 target 0

eui.50802714000000000060e82827140009 : NVMe TCP Disk (eui.50802714000000000060e82827140009)
  vmhba64:C0:T0:L9 LUN:9 state:active Local HBA vmhba64 channel 0 target 0
  vmhba65:C0:T0:L9 LUN:9 state:active Local HBA vmhba65 channel 0 target 0
```

ESXi Path Selection Policy with the High-Performance Plug-in

VMware introduced a new multipath plugin in ESXi 7.0 for high-speed devices such as NVMe. The High Performance Plugin (HPP) replaces NMP (Native Multipath Plugin) for NVMe over TCP devices as the default plugin. Unlike the Path Selection Policy (PSP) used by NMP, HPP uses Path Selection Schemes (PSS) to select paths for IO devices. By default, LB-RR (Load Balance – Round Robin) is selected as the PSS by HPP; however, this can be changed to FIXED or another LB PSS using `esxcli` commands.

For more information on the VMware HPP, see:

<https://docs.vmware.com/en/VMware-vSphere/8.0/vsphere-storage/GUID-F7B60A5A-D077-4E37-8CA7-8CB912173D24.html>

For more information on the ESXi host for NVMe/TCP devices, run the following `esxcli` commands:

```
# esxcli storage hpp device list
```

```
eui.50802714000000000060e82827140000
Device Display Name: NVMe TCP Disk (eui.50802714000000000060e82827140000)
Path Selection Scheme: LB-RR
Path Selection Scheme Config: {iops=1000,bytes=10485760;}
Current Path: vmhba64:C0:T0:L0
Working Path Set: vmhba64:C0:T0:L0, vmhba65:C0:T0:L0
Is SSD: true
Is Local: false
Paths: vmhba64:C0:T0:L0, vmhba65:C0:T0:L0
Use ANO: false
```

```
eui.50802714000000000060e82827140001
Device Display Name: NVMe TCP Disk (eui.50802714000000000060e82827140001)
Path Selection Scheme: LB-RR
Path Selection Scheme Config: {iops=1000,bytes=10485760;}
Current Path: vmhba64:C0:T0:L1
Working Path Set: vmhba64:C0:T0:L1, vmhba65:C0:T0:L1
Is SSD: true
Is Local: false
Paths: vmhba64:C0:T0:L1, vmhba65:C0:T0:L1
Use ANO: false
```

```
# esxcli storage hpp device list
```

```
tcp.vmn4:b8:3f:d2:2a:0b:10-tcp.192.168.10.10:4420-eui.50802714000000000060e82827140000
Runtime Name: vmhba64:C0:T0:L0
Device: eui.50802714000000000060e82827140000
Device Display Name: NVMe TCP Disk (eui.50802714000000000060e82827140000)
Path State: active
Path Config:
```

```
tcp.vmn4:b8:3f:d2:2a:0b:10-tcp.192.168.10.10:4420-eui.50802714000000000060e82827140001
Runtime Name: vmhba64:C0:T0:L1
Device: eui.50802714000000000060e82827140001
Device Display Name: NVMe TCP Disk (eui.50802714000000000060e82827140001)
Path State: active
Path Config:
```

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